money) and inflation. Therefore, with the help of coordinate the value of money can also be measured. The value of money (purchasing power) decreases with an increase in the price level, while the value of money (purchasing power) increases with a fall in the price level.

84. The Sixth and the Eight Five Year Plans covered the Period 1980-1985 and 1992-1997 respectively. The Seventh Five Year Plan covered the period:

(a) 1987-1992 (b) 1986-1991
(c) 1985-1990 (d) 1988-1994

Ans. (c) : The time of the Seventh Five Year Plan was from 1st April, 1985 to 31st March, 1990 AD. After this the Eighth Five Year Plan which was to start from 1st April, 1990 could not be started on time due to political changes in the center in two years. Thus the time of the Eighth Five Year Plan was from 1st April, 1992 to 31st March, 1997 AD.

85. Schemes of (i) Urban Micro-Enterprises, (ii) Urban Wage Employment and (iii) Housing and Shelter Up-gradation are parts of:

(a) Integrated Rural Development Programme
(b) Nehru Rojgar Yojana
(c) Jawahar Rojgar Yojana
(d) Prime Minister's Rojgar Yojana

Ans. (b) : The Nehru Rozgar Yojana (1989) has been designed to provide employment to the urban unemployed and under-employed poor.

86. Given below are two statements one labelled as Assertion (A) and the other labelled as (Reason):

Assertion (A) : The emergence of economic globalism does not imply the decline of socialist ideology.

Reason (R) : The ideology of socialism believes in Universalism and globalism.

In the context of the above two statements, which one of the following is correct?

(a) Both A and R are true and R is the correct explanation of A
(b) Both A and Ra re true but R is not correct explanation of A
(c) A is true but R is false.
(d) A is false but R is true.

Ans. (b) : Globalism is the policy of placing the interests of the world above those of individual nations. This also means that there would be free movement of goods and services across national boundaries. Socialism is an economic system based on state ownership of means of production including capital. Universalism means application of certain basic principles at universal level.

87. The contribution of India's small scale sector to the gross turnover in the manufacturing sector since 1992 has been of the order of:

(a) 40% (b) 34%
(c) 30% (d) 45%

Ans. (a) : The contribution of small scale industries to employment and exports is substantial. The sector contributes over 40 per cent of the gross turnover in the manufacturing sector, 45 per cent of manufactured exports and 25 per cent of total exports. As in 2015.

88. Which of the following come under Non-plan expenditure?

1. Subsidies
2. Interest payments
3. Defence expenditure
4. Maintenance expenditure for the infrastructure created in the previous plans

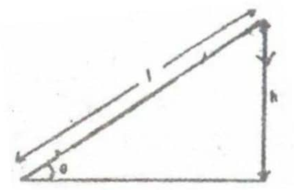
Choose the correct answer using the codes given below:

(a) 1 and 2 (b) 1 and 3
(c) 2 and 4 (d) 1, 2, 3 and 4

Ans. (d) : Non-plan expenditure is what the government spends on the so-called non productive areas, such as salaries, subsidies, loan and interest payments, defence expenditure and maintenance expenditure for the infrastructure created in the previous plans. So all the given codes are correct and thus option (d) is correct.

89. National Income is the:

(a) Net National Product at market price
(b) Net National Product at factor cost
(c) Net Domestic Product at market price
(d) Net Domestic Product at factor cost



Ans. (b) : National Income of countries is the total amount of income that is accrued by it through various economic activities in one year. It is used in determining the progress of the country.
It is also defined as Net National Product at factor cost.
National Income = Net National Product + Subsidies – Interest Taxes.

90. The Minimum Alternative Tax (M.A.T.) was introduced in the Budget of the Government of India for the year:

(a) 1991-92 (b) 1992-93
(c) 1995-96 (d) 1996-97

Ans. (d) : Minimum Alternate Tax (MAT) is applied when the taxable income calculated according to the I-T Act provisions is found to be less than 15.5 percent of the book profit under the companies Act. It was introduced by the finance act, 1996.

PHYSICS

91. The tendency of a liquid drop to contract and occupy minimum area is due to:

(a) Surface tension (b) Viscosity
(c) Density (d) vapour pressure

Ans. (a) : The surface area of a sphere is minimum for a given volume. Therefore, from the spherical shape of the liquid droplets, it can be concluded that the liquid wants to keep its surface area minimum. Thus it is clear that there is a kind of tension in the surface of the liquid which is called surface tension. The surface tension of a liquid is the force that acts F on the unit length of an imaginary line drawn on the surface of the fluid in the plane of the surface and perpendicular to the line. Its unit is Newton/Meter.
Surface tension T = F /L
where, F= force, L = length of line.

92. Which one of the following is a vector quantity?

(a) Momentum (b) Pressure
(c) Energy (d) Work

Ans. (a) : Scalars quantity are quantities that have magnitude only and they are independent of direction. Vectors quantity have both magnitude and direction. Momentum is the product of the mass and velocity of an object (p = mv). Momentum is a vector quantity, since it has a direction as well as a magnitude. The rest of quantities in option pressure, work and energy have magnitude but not direction.

93. A smooth inclined at an angle q with the horizontal as shown in the above figure. A body starts from rest and slides down the inclined surface. The time taken by the body to reach the bottom is:

(a) $\sqrt{\frac{2h}{g}}$ (b) $\sqrt{\frac{2l}{g}}$
(c) $\frac{1}{\sin\theta}\sqrt{\frac{2h}{g}}$ (d) $\sin\theta\sqrt{\frac{2h}{g}}$

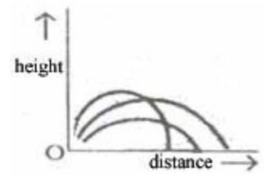
Ans. (c) :
Component of acceleration along the inclined plane will be gsinθ
$l = g\sin\theta \times t^2/2$
$\sin\theta = h/l \Rightarrow l = \sin\theta / h$
$t = \sin\theta\sqrt{\frac{2h}{g}}$

94. A girl is swinging on a swing in sitting position. When the same girl stands up, the period of swing will:

(a) Be shorter
(b) Be longer
(c) Depend on the height of the girl
(d) Not change

Ans. (a) : A swing is a example SHM,
Time period for it will be :
$T = 2\pi\sqrt{\frac{\rho}{g}}$
$T \propto \sqrt{l}$
When girl stands up, centre of mass of her body will go up, so will be less then 'T'.

95. A body standing at the point 'O' in the given diagram, throws a ball three times with the same force, but projecting it along different inclinations from the ground. The results of the throws have been plotted in the diagram. Which one of the following is a valid conclusion?



(a) The larger the initial inclination, the longer the throw
(b) The larger the height reached, the longer the throw
(c) The larger the height reached, the shorter the throw
(d) The larger the initial inclination, the greater the height reached

Ans. (c) : In given figure ball is shown thrice with same force but with different inclination.
Horizontal range = v cosθ
Vertical range = v sinθ
If body is projected with higher altitude, it means value of 'θ' is greater than 45°
In such situation, horizontal range will be lesser because it depends upon value of cosθ.
As value of cosθ for angle greater than 45° is lesser compare to sinθ so, horizontal range (throw) will be lesser.

96. The working principle of a washing machine: